MATHEMATICS

SET & RELATION

THEORY SHEET

ENGLISH

SET & RELATION

1. SET THEORY

Well defined collection of objects which are distinct and distinguishable. A collection is said to be well defined if each and every element of the collection has some definition.

1.1 Notation of a set : Sets are denoted by capital letters like A, B, C or { } and the entries within the bracket are known as elements of set.

1.2 Cardinal number of a set : Cardinal number of a set X is the number of elements of a set X and it is denoted by n(X) or |X| e.g. $X = \{x_1, x_2, x_3\} \therefore n(X) = 3$

2. REPRESENTATION OF SETS

2.1 Set Listing Method (Roster Method) :

In this method a set is described by listing all the elements, separated by commas, within braces { }

2.2 Set builder Method (Set Rule Method) :

In this method, a set is described by characterizing property P(x) of its elements x. In such case the set is described by {x : P(x) holds} or {x | P(x) holds}, which is read as the set of all x such that P(x) holds. The symbol '|' or '**:**' is read as such that.

3. TYPE OF SETS

3.1 Finite set :

A set X is called a finite set if its element can be listed by counting or labeling with the help of natural numbers and the process terminates at a certain natural number n. i.e. n(X) = finite no. eg (a) A set of English Alphabets (b) Set of soldiers in Indian Army

3.2 Infinite set :

A set whose elements cannot be listed counted by the natural numbers (1, 2, 3.....n) for any number n, is called a infinite set. e.g.

(a) A set of all points in a plane (b) $X = \{x : x \in R, 0 < x < 0.0001\}$ (c) $X = \{x : x \in Q, 0 \le x \le 0.0001\}$

3.3 Singleton set :

A set consisting of a single element is called a singleton set. i.e. n (X) = 1,

e.g. $\{x : x \in N, 1 < x < 3\}$, {{}} : Set of null set, {ϕ} is a set containing alphabet ϕ.

3.4 Null set :

A set is said to be empty, void or null set if it has no element in it, and it is denoted by ϕ. i.e. X is a null set if n(X) = 0. e.g. : $\{x : x \in R \text{ and } x^2 + 2 = 0\}$, $\{x : x > 1 \text{ but } x < 1/2\}$, $\{x : x \in R, x^2 < 0\}$

3.5 Equivalent Set :

Two finite sets A and B are equivalent if their cardinal numbers are same i.e. n(A) = n(B).

3.6 Equal Set :

Two sets A and B are said to be equal if every element of A is a member of B and every element of B is a member of A. i.e. A = B, if A and B are equal and $A \neq B$, if they are not equal.

4. SUBSET

A set A is said to be a subset of B if all the elements of A are present in B and is denoted by $A \subseteq B$ (read as A is subset of B) and symbolically written as : $x \in A \Rightarrow x \in B \Leftrightarrow A \subseteq B$

4.1 Number of subsets :

Consider a set X containing n elements as $\{x_1, x_2, ..., x_n\}$ then the total number of subsets of $X = 2^n$

Proof : Number of subsets of above set is equal to the number of selections of elements taking any number of them at a time out of the total n elements and it is equal to 2^n

$\because \quad {}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_n = 2^n$

4.2 Types of Subsets :

A set A is said to be a **proper subset** of a set B if every element of A is an element of B and B has at least one element which is not an element of A and is denoted by $A \subset B$.

The statement $A \subset B$ can be written as $B \supset A$, then B is called the **super set** of A and is written as $B \supset A$.

4.3 Power Set

The collection of all subsets of set A is called the power set of A and is denoted by P(A)

i.e. $P(A) = \{x : x \text{ is a subset of } A\}$. If $X = \{x_1, x_2, x_3, ...x_n\}$ then $n(P(X)) = 2^n ; n(P(P(x))) = 2^{2^n}$.

Ex.1 If a set $A = \{a, b, c\}$ then find the number of subsets of the set A and also mention the power set of all the subsets of A.

Sol. Since $n(A) = 3$

$\therefore$ number of subsets of A is $2^3 = 8$

and set of all those subsets is P(A) named as power set

$P(A) : \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}\}$

Ex.2 Show that $n\{P\{P[P(\phi)]\}\} = 4$

Sol. We have $P(\phi) = \{\phi\} \quad \therefore \quad P(P(\phi)) = \{\phi, \{\phi\}\}$

$\Rightarrow \quad P[P(P(\phi))] = \{ \phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$.

Hence, $n\{P\{P[P(\phi)]\}\} = 4$

5. OPERATIONS ON SETS & VENN DIAGRAM

5.1 Universal Set

It is a set which includes all the sets under considerations i.e. it is a super set of each of the given set. Thus, a set that contains all sets in a given context is called the universal set. It is denoted by U. e.g. If $A = \{1, 2, 3\}$, $B = \{2, 4, 5, 6\}$ and $C = \{1, 3, 5, 7\}$, then $U = \{1, 2, 3, 4, 5, 6, 7\}$ can be taken as the universal set.

5.2 complementary set

Complementary set of a set A is a set containing all those elements of universal set which are not in A. It is denoted

by $\overline{A}$, A^C or A'. So $A^C = \{x : x \in U \text{ but } x \notin A\}$. e.g. If set $A = \{1, 2, 3, 4, 5\}$ and universal set $U = \{1, 2, 3, 4,.....50\}$ then $\overline{A} = \{6, 7, ...50\}$

5.3 Union of sets :

If A and B are two sets then union ($\cup$) of A and B is the set of all those elements which belong either to A or to B or to both A and B. It is also defined as $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

5.4 Intersection of sets :

If A and B are two sets then intersection ($\cap$) of A and B is the set of all those elements which belong to both A and B. It is also defined as $A \cap B = \{x : x \in A \text{ and } x \in B\}$

5.5 Difference of two sets :

If A and B are two sets then the difference of A and B, is the set of all those elements of A which do not belong to B.

Thus, $A - B = \{x : x \in A \text{ and } x \notin B\}$

or $A - B = \{x \in A ; x \notin B\}$

Clearly $x \in A - B \Leftrightarrow x \in A$ and $x \notin B$

5.6 Symmetric difference of two sets :

Set of those elements which are obtained by taking the union of the difference of A & B is (A – B) & the difference of B & A is (B – A), is known as the symmetric difference of two sets A & B and it is denoted by ($A \Delta B$). Thus $A \Delta B = (A - B) \cup (B - A)$

5.7 Disjoint Set

Sets A and B are said to be disjoint iff A and B have no common element or $A \cap B = \phi$. If $A \cap B \neq \phi$ then A and B are said to be intersecting or overlapping sets.

e.g. : (i) If $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$ and $C = \{4, 7, 9\}$ then A and B are disjoint set where B and C are intersecting sets. **(ii)** Set of even natural numbers and odd natural numbers are disjoint sets.

5.8 Number of elements in different sets

If A, B & C are finite sets and U be the finite universal set, then

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cup B) = n(A) + n(B)$ (if A & B are disjoint sets)

(iii) $n(A - B) = n(A) - n(A \cap B)$

(iv) $n(A \Delta B) = n[(A - B) \cup (B - A)] = n(A) + n(B) - 2n(A \cap B)$

(v) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

(vi) $n(A' \cup B') = n(A \cap B)' = n(U) - n(A \cap B)$

(vii) $n(A' \cap B') = n(A \cup B)' = n(U) - n(A \cup B)$

Ex.1 If $A = \{x : x = 2n + 1, n \in Z$ and $B = \{x : x = 2n, n \in Z\}$, then find $A \cup B$.

Sol. $A \bigcup B = \{x : x \text{ is an odd integer}\} \cup \{x : x \text{ is an even integer}\} = \{x : x \text{ is an integer}\} = Z$

Ex.2 If $A = \{x : x = 3n, n \in Z\}$ and

$B = \{x : x = 4n, n \in Z\}$ then find $A \cap B$.

Sol. We have,

$x \in A \cap B \Leftrightarrow x = 3n, n \in Z$ and $x = 4n, n \in Z$

$\Leftrightarrow x$ is a multiple of 3 and x is a multiple of 4

$\Leftrightarrow x$ is a multiple of 3 and 4 both

$\Leftrightarrow x$ is a multiple of 12 $\Leftrightarrow x = 12n, n \in Z$

Hence $A \cap B = \{x : x = 12n, n \in Z\}$

Ex.3 If A and B be two sets containing 3 and 6 elements respectively, what can be the minimum number of elements in $A \cup B$? Find also, the maximum number of elements in $A \cup B$.

Sol. We have, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

This shows that $n(A \cup B)$ is minimum or maximum according as $n(A \cap B)$ is maximum or minimum respectively.

Case-I

When $n(A \cap B)$ is minimum, i.e., $n(A \cap B) = 0$

This is possible only when $A \cap B = \phi$. In this case,

$n(A \cup B) = n(A) + n(B) - 0 = n(A) + n(B) = 3 + 6 = 9$.

So, maximum number of elements in $A \cup B$ is 9.

Case-II

When $n(A \cap B)$ is maximum.

This is possible only when $A \subseteq B$. In this case, $n(A \cap B) = 3$

$\therefore \quad n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= (3 + 6 - 3) = 6$

So, minimum number of elements in $A \cup B$ is 6.

Ex.4 If $A = \{2, 3, 4, 5, 6, 7\}$ and $B = \{3, 5, 7, 9, 11, 13\}$ then find $A - B$ and $B - A$..

Sol. $A - B = \{2, 4, 6\}$ & $B - A = \{9, 11, 13)$

Ex.5 In a class of 140 students numbered 1 to 140, all even numbered students opted Mathematics course, those whose number is divisible by 3 opted physics course and those whose number is divisible by 5 opted Chemistry course. Then the number of students who did not opt for any of the three courses is :

(1) 1 (2) 42 (3) 38 (4) 102

140 विद्यार्थियों, जिनके क्रमांक 1 से 140 हैं, की एक कक्षा में सभी सम क्रमांक के विद्यार्थियों ने गणित विषय चुना है, उन्होंने जिनके क्रमांक 3 से विभाजित होते है भौतिक शास्त्र विषय चुना है तथा उन्होंने जिनके क्रमांक 5 से विभाजित होते हैं, रसायन शास्त्र विषय चुना है। तो उन विद्यार्थियों की संख्या, जिन्होंने इन तीन में से कोई भी विषय नहीं चुना है, हैं :

(1) 1 (2) 42 (3) 38 (4) 102

A. 3

Sol. $n(A)$ = Number of students opted Mathematics = 70

$n(B)$ = Number of students opted Physics = 46

$n(C)$ = Number of students opted Chemistry = 28

$n(A \cap B) = 23$

$n(B \cap C) = 9$

$n(A \cap C) = 14$

$n(A \cap B \cap C) = 4$

$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$

$= (70 + 46 + 28) - (23 + 9 + 14) + 4$

$= 102$

No of students did not opt any of the three Sujbect = 140 – 102

= 38

Ex.6 Let A, B and C be sets such that $\phi \neq A \cap B \subseteq C$. Then which of the following statements is not true

(1) $B \cap C \neq \phi$ (2) $(C \cup A) \cap (C \cup B) = C$

(3) If $(A - C) \subseteq B$, then $A \subseteq B$ (4) If $(A - B) \subseteq C$, then $A \subseteq C$

माना समुच्चय A, B तथा C इस प्रकार हैं कि $\phi \neq A \cap B \subseteq C$, तो निम्न में से कौनसा कथन सत्य नहीं है?

(1) $B \cap C \neq \phi$ (2) $(C \cup A) \cap (C \cup B) = C$

(3) यदि $(A - C) \subseteq B$, तो $A \subseteq B$ (4) यदि $(A - B) \subseteq C$, तो $A \subseteq C$

A. 3

sol. $\because A \cap B \subseteq C$ and $A \cap B \neq \phi$

(1) $B \cap C \neq \phi$ is correct

(2) $(C \cup A) \cap (C \cup B) = C \cup (A \cap B) = C$ (correct) (becasue $A \cap B \subseteq C$)

(3) If $A = C$ then $A - C = \phi$ Clearly $\phi \subseteq B$ but $A \subseteq B$ is not always true.

(4) $\because A - B \subseteq C$ and $A \cap B \subseteq C$ so $A \subseteq C$ (correct)

Ex.7 Let $X = \{n \in N : 1 \leq n \leq 50\}$. If $A = \{n \in X : n$ is a multiple of $2\}$ and $B = \{n \in X : n$ is a multiple of $7\}$, then the number of elements in the smallest subset of X containing both A and B is_____ .

माना $X = \{n \in N : 1 \leq n \leq 50\}$. यदि $A = \{n \in X : n$, 2 का एक गुणज है$\}$ तथा $B = \{n \in X : n$, 7 का एक गुणज है$\}$, तो X के सबसे छोटे उपसमुच्चय, जिसमें A तथा B दोनों हैं, में अवयवों की संख्या है_____।

A. 29

Question ID : 4050361314

Sol. A = {2, 4, 6.....50}

B = {7, 14, 21 .....49}

Answer $= n(A \cup B)$

$= n(A) + n(B) - n(A \cap B)$

$= 25 + 7 - 3$

$= 29$

Ex.8 A survey shows that 63% of the people in a city read newspaper A whereas 76% read newspaper B. If x% of the people read both the newspapers, then a possible value of x can be :

एक सर्वेक्षण दर्शाता है कि एक शहर में 63% लोग समाचार पत्र A पढ़ते हैं, 76% समाचार पत्र B पढ़ते हैं। यदि x% लोग दोनों समाचार पत्र पढ़ते हैं, तो x का एक संभावित मान हो सकता है :

(1) 65 (2) 29 (3) 37 (4) 55

Ans (4)

Sol. A = Person who read newspaper of type A

B = Person who read newspaper of type B

$n(A) = 63\%$

$n(B) = 76\%$

$n(A \cap B) = x\%$

Let $n(U) = 100$

$n(A) = 63, n(B) = 76, n(A \cap B) = x$

$n(A \cup B) = n(A) + n(B) - n(A \cap B) \le 100$

$63 + 76 - x \le 100$

$x \ge 39$

$n(A \cap B) \le \min\{n(A), n(B)\}$

$\therefore 39 \le x \le 63$

Ex.9 Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^{n} Y_i = T$, where each X_i contains 10 elements and each Y_i contains 5 elements. If each element of the set T is an element of exactly 20 of sets X_i's and exactly 6 of sets Y_i's, then n is equal to :

माना $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^{n} Y_i = T$ है, जहाँ प्रत्येक X_i में 10 अवयव हैं तथा प्रत्येक Y_i में 5 अवयवव हैं। यदि T का प्रत्येक अवयव ठीक 20, X_i समुच्चयों का एक अवयव है तथा ठीक 6, Y_i समुच्चयों का एक अवयव है, तो n का मान है :

(1) 30 (2) 50 (3) 15 (4) 45

Ans. (1)

Sol. Number of elements in $T = \frac{50 \times 10}{20} = \frac{n \times 5}{6}$

$\Rightarrow n = 30$

6. ALGEBRAIC OPERATIONS ON SETS

6.1 Idempotent operation :

For any set A, we have (i) $A \cup A = A$ and (ii) $A \cap A = A$

6.2 Identity operation :

For any set A, we have

(i) $A \cup \phi = A$ and

(ii) $A \cap U = A$ i.e. ϕ and U are identity elements for union and intersection respectively

6.3 Commutative operation :

For any set A and B, we have

(i) $A \cup B = B \cup A$ and (ii) $A \cap B = B \cap A$

i.e. union and intersection are commutative.

6.4 Associative operation :

If A, B and C are any three sets then

(i) $(A \cup B) \cup C = A \cup (B \cup C)$

(ii) $(A \cap B) \cap C = A \cap (B \cap C)$

i.e. union and intersection are associative.

6.5 Distributive operation :

If A, B and C are any three sets then

(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

i.e. union and intersection are distributive over intersection and union respectively.

6.6 De-Morgan's Principle :

If A and B are any two sets, then

(i) $(A \cup B)' = A' \cap B'$

(ii) $(A \cap B)' = A' \cup B'$

7. CARTESIAN PRODUCT OF TWO SETS

Cartesian product of A to B is a set containing the elements in the form of ordered pair (a, b) such that $a \in A$ and $b \in B$. It is denoted by $A \times B$.

i.e. $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$

$= \{(a_1, b_1),(a_1, b_2),(a_2, b_1),(a_2, b_2),(a_3, b_1),(a_3, b_2)\}$

If set $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2\}$ then

$A \times B$ and $B \times A$ can be written as :

$A \times B = \{(a, b) : a \in A \text{ and } b \in B \text{ and}$

$B \times A = \{(b, a) ; b \in B \text{ and } a \in A\}$

Clearly $A \times B \neq B \times A$ until A and B are equal

Note :

1. If number of elements in A : $n(A) = m$ and $n(B) = n$ then number of elements in $(A \times B) = m \times n$

Ex.1 If the number of elements in A is m and number of element in B is n then find

(i) The number of elements in the power set of $A \times B$.

(ii) number of relation defined from A to B

Sol. **(i)** Since $n(A) = m$; $n(B) = n$ then $n(A \times B) = mn$

So number of subsets of $A \times B = 2^{mn}$

$\Rightarrow \quad n(P(A \times B)) = 2^{mn}$

(ii) number of relation defined from A to $B = 2^{mn}$

Any relation which can be defined from set A to set B will be subset of $A \times B$

$\because$ $A \times B$ is largest possible relation $A \to B$

$\therefore$ no. of relation from $A \to B$ = no. of subsets of set $(A \times B)$

Ex.2 Let A and B be two non-empty sets having elements in common, then prove that $A \times B$ and $B \times A$ have n^2 elements in common.

Sol. We have $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$

On replacing C by B and D by A, we get

$\Rightarrow \quad (A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$

It is given that AB has n elements so $(A \cap B) \times (B \cap A)$ has n^2 elements

But $(A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$

$\therefore$ $(A \times B) \cap (B \times A)$ has n^2 elements

Hence A × B and B × A have n^2 elements in common.

Ex.3 Let Z be the set of integers. If $A=\left\{x\in Z:2^{(x+2)\left(x^2-5x+6\right)}=1\right\}$ and $B=\{x\in Z:-3<2x-1<9\}$, then the number of subsets of the set A × B, is :

(1) 2^{15} (2) 2^{10} (3) 2^{18} (4) 2^{12}

माना Z पूर्णाकों का समुच्चय है। यदि $A=\left\{x\in Z:2^{(x+2)\left(x^2-5x+6\right)}=1\right\}$ तथा $B=\{x\in Z:-3<2x-1<9\}$ तो A × B के उपसमुच्चयों की संख्या है –

(1) 2^{15} (2) 2^{10} (3) 2^{18} (4) 2^{12}

A. 1

sol. $A=\left\{x\in Z:2^{(x+2)\left(x^2-5x+6\right)}=1\right\}$

$(x+2)(x^2-5x+6)=0$

$(x+2)(x-2)(x-3)=0$

$x=-2,\ x=2$ and $x=3$

$B=\{x\in Z:-3<2x-1<9\}$

$-3<2x-1<9$

$-2<2x<10$

$-1<x<5$

x (1, 2, 3, 4)

A ∈ (2, 2, 3)

B ∈ (0, 1, 2, 3, 4)

To no. of element in A × B = 15

So subset = 2^{15}

8. RELATION

A relation R from set X to Y ($R : X \to Y$) is a correspondence between set X to set Y by which some or more elements of X are associated with some or more elements of Y. Therefore a relation (or binary relation) R, from a non-empty set X to another non-empty set Y, is a subset of X × Y. i.e. $R_H : X \to Y$ is nothing but subset of A × B.

Note :

(i) If a is related to b then symbolically it is written as a R b where a is pre-image and b is image

(ii) If a is not related to b then symbolically it is written as a $\not R$ b.

8.1 Domain, Co-domain & Range of Relation :

Domain : of relation is collection of elements of the first set which are participating in the correspondence i.e. it is set of all pre-images under the relation R.

Co-Domain : All elements of set Y irrespective of whether they are related with any element of X or not constitute co-domain.

Range : of relation is a set of those elements of set Y which are participating in correspondence i.e. set of all images.

Ex.1 Let R be the relation on the set N of natural numbers defined by

$R : \{(x, y)\} : x + 3y = 12 \ \ x \in N, y \in N\}$ Find

(i) R **(ii)** Domain of R **(iii)** Range of R

Sol. **(i)** We have, $x + 3y = 12 \Rightarrow x = 12 - 3y$

Putting $y = 1, 2, 3$, we get $x = 9, 6, 3$ respectively

For $y = 4$, we get $x = 0 \notin N$. Also for $y > 4$, $x \notin N$

$\therefore$ $R = \{(9, 1), (6, 2), (3, 3)\}$

(ii) Domain of $R = \{9, 6, 3\}$

(iii) Range of $R = \{1, 2, 3\}$

9. TYPES OF RELATIONS

9.1 Reflexive Relation

$R : X \to Y$ is said to be reflexive iff $x \, R \, x \ \forall \ x \in X$. i.e. every element in set X, must be a related to itself therefore $\forall \ x \in X; (x, x) \in R$ then relation R is called as reflexive relation.

9.2 Identity Relation :

Let X be a set. Then the relation $I_x = \{(x, x) : x \in X\}$ on X is called the identity relation on X. i.e. a relation I_x on X is identity relation if every element of X related to itself only. e.g. $y = x$

Note : All identity relations are reflexive but all reflexive relations are not identity.

9.3 Symmetric Relation

$R : X \to Y$ is said to be symmetric if $f(x, y) \in R \Rightarrow (y, x) \in R$ for all $(x, y) \in R$ i.e. $x \, R \, y \Rightarrow y \, R \, x$ for all $(x, y) \in R$. e.g. perpendicularity of lines in a plane is symmetric relation.

9.4 Transitive Relation

$R : X \to Y$ is transitive iff $(x, y) \in R$ and $(y, z) \in R \Rightarrow (x, z) \in R$ for all (x, y) and $(y, z) \in R$. i.e. $x \, R \, y$ and $y \, R \, z \Rightarrow x \, R \, z$. e.g. The relation "being sister of" among the members of a family is always transitive.

Note :

(i) Every null relation is a symmetric and transitive relation.

(ii) Every singleton relation is a transitive relation.

(iii) Universal and identity relations are reflexive, symmetric as well as transitive.

9.5 Anti-symmetric Relation

Let A be any set. A relation R on set A is said to be an antisymmetric relation iff $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$ for all $a, b \in A$ e.g. Relations "being subset of"; "is greater than or equal to" and "identity relation on any set A" are antisymmetric relations.

9.6 Equivalence Relation

A relation R from a set X to set Y ($R : X \to Y$) is said to be an equivalence relation iff it is reflexive, symmetric as well as transitive. The equivalence relation is denoted by ~ e.g. Relation "is equal to" Equality, Similarity and congruence of triangles, parallelism of lines are equivalence relation.

Ex.1 If $X=\{x_1, x_2, x_3\}$ and $y = (x_1, x_2, x_3, x_4, x_5\}$ then find which is a reflexive relation of the following :

(a) $R_1 : \{(x_1, x_1), (x_2, x_2)$

(b) $R_1 : \{(x_1, x_1), (x_2, x_2), (x_3, x_3)$

(c) $R_3 : \{(x_1, x_1), (x_2, x_2), (x_3, x_3), (x_1, x_3), (x_2, x_4)$ (d) $R_3 : \{(x_1, x_1), (x_2, x_2), (x_3, x_3), (x_4, x_4)$

Sol. (a) non-reflexive because $(x_3, x_3) \notin R_1$

(b) Reflexive

(c) Reflexive

(d) non-reflexive because $x_4 \notin X$

Ex.2 If x = {a, b, c} and y = {a, b, c, d, e, f} then find which of the following relation is symmetric relation :

$R_1 : \{ \}$ i.e. void relation

$R_2 : \{(a, b)\}$

$R_3 : \{(a, b), (b, a)(a, c)(c, a)(a, a)\}$

Sol. R_1 is symmetric relation because it has no element in it.

R_2 is not symmetric because $(b, a) \in R_2$ & R_3 is symmetric.

Ex.3 If x = {a, b, c} and y = (a, b, c, d, e} then which of the following are transitive relation.

(a) $R_1 = \{ \}$

(b) $R_2 = \{(a, a)\}$

(c) $R_3 = \{(a, a\}.(c, d)\}$

(d) $R_4 = \{(a, b), (b, c)(a, c), (a, a), (c, a)\}$

Sol. **(a)** R_1 is transitive relation because it is null relation.

(b) R_2 is transitive relation because all singleton relations are transitive.

(c) R_3 is transitive relation

(d) R_4 is not a transitive relation

Ex.4 Let R be a relation on the set N of natural numbers defined by xRy $\Leftrightarrow$ x divides y ′ for all x, y $\in$ N.

Sol. This relation is an antisymmetric relation on set N. Since for any two numbers a, b $\in$ N. a | b and b | a $\Rightarrow$ a = b, i.e., a R b and b R a $\Rightarrow$ a = b.

It should be noted that this relation is not antisymmetric on the set Z of integers, because we find that for any non zero integer a a R (a) and (–a) R a, but a ≠ – a.

Ex.5 Prove that the relation R on the set Z of all integers numbers defined by (x, y) $\in$ R $\Leftrightarrow$ x – y is divisible by n is an

equivalence relation on Z.

Sol. We observe the following properties

Reflexivity :

For any $a \in N$, we have

$a - a = 0 \times n \Rightarrow a - a$ is divisible by $n \Rightarrow (a, a) \in R$

Thus (a, a) R for all Z. so, R is reflexive on Z.

Symmetry :

Let $(a, b) \in R$. Then $(a, b) \in R \Rightarrow (a - b)$ is divisible by n

$\Rightarrow (a - b) = np$ for some $p \in Z \Rightarrow b - a = n(-p)$

$\Rightarrow b - a$ is divisble by n $\Rightarrow (b, a) \in R$

Thus $(a, b) \in R \Rightarrow (b, a) \in R$ for all a, b, $\in Z$.

So R is symmetric on Z.

Transitivity :

Let $a, b, c \in Z$ such that $(a, b) \in R$ and $(b, c) \in R$. Then

$(a, b) \in R \Rightarrow (a - b)$ is divisible by $n \Rightarrow a - b = np$ for some $p \in Z$

$(b, c) \in R \Rightarrow (b - c)$ is divisible by $n \Rightarrow b - c = np$ for some $q \in Z$

$\therefore$ $(a, b) \in R$ and $(b, c) \in R \Rightarrow a - b = np$ and $b - c = nq$

$\Rightarrow (a - b) + (b - c) = np + nq \Rightarrow a - c = n(p + q)$

$\Rightarrow$ $a - c$ is divisible by $n \Rightarrow (a, c) \in R$

Thus $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in Z$. So R is transitive relation on Z.

Thus, R being reflexive, symmetric and transitive is an equivalence relation on Z.

Ex.6 Let a relation R_1 on the set R of real numbers be defined as $(a, b) \in R_1 \Leftrightarrow 1 + ab > 0$ for all $a, b \in R$. Show that R_1 is reflexive and symmetric but not transitive.

Sol. We observe the following properties :

Reflexivity :

Let a be an arbitrary element of R. Then

$a \in R \Rightarrow 1 + a \,.\, a = 1 + a^2 > 0 \Rightarrow (a, a) \in R_1$

Thus $(a, a) \in R_1$ for all $a \in R$. So R_1 is reflexive on R.

Symmetry :

Let $(a, b) \in R$. Then

$(a, b) \in R_1 \Rightarrow 1 + ab > 0 \Rightarrow 1 + ba > 0$

$\Rightarrow (b, a) \in R_1$

Thus $(a, b) \in R_1 \Rightarrow (b, a) \in R_1$ for all $a, b \in R$

So R_1 is symmetric on R

Transitive :

We observe that $(1, 1/2) \in R_1$ and $(1/2, -1) \in R_1$ but $(1,-1) \in R_1$ because $1 + 1 \times (-1) = 0 \neq 0$. So R_1 is not transitive on R.

Ex.7 Let A be the set of first ten natural numbers and let R be a relation on A defined by $(x, y) \in R \Leftrightarrow x + 2y = 10$ i.e., $R = \{(x, y) : x \in A, y \in A \text{ and } x + 2y = 10\}$. Express R and R^{-1} as sets of ordered pairs. Determine also :

(i) Domains of R and R^{-1}

(ii) Range of R and R^{-1}

Sol. We have $(x, y) \in R \Leftrightarrow x + 2y = 10 \Leftrightarrow y = \frac{10 - x}{2}$, $x, y \in A$

where $A = \{1,2,3,4,5,6,7,8,9,10\}$

Now, $x = 1 \Rightarrow y = \frac{10-1}{2} = \frac{9}{2} \notin A$

This shows that 1 is not related to any element in A. Similarly we can observe that 3, 5, 7, 9 and 10 are not related to any element of a under the defined relation. Further we find that

for $x = 2$, $y = \frac{10-2}{2} = 4 \in A \quad \therefore (2, 4) \in R$

for $x = 4$, $y = \frac{10-4}{2} = 3 \in A \quad \therefore (4, 3) \in R$

for $x = 6$, $y = \frac{10-6}{2} = 2 \in A \quad \therefore (6, 2) \in R$

for $x = 8$, $y = \frac{10-8}{2} = 1 \in A \quad \therefore (8, 1) \in R$

Thus $R = \{(2, 4), (4, 3), (6, 2), (8, 1)\}$

$\Rightarrow R^{-1} = \{(4, 2), (3, 4), (2, 6), (1, 8)\}$

Clearly, Dom (R) = $\{2, 4, 6, 8\}$ = Range (R^{-1})

and Range (R) = $\{4, 3, 2, 1\}$ = Dom (R^{-1})

Ex.8 Let $A = \{-2, -1, 0, 1, 2, 3\}$. Let R be a relation on A defined by xRy if and only if $y = \max\{x, 1\}$. Let l be the number of elements in R. Let m and n be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then $l + m + n$ is equal to :

माना $A = \{-2, -1, 0, 1, 2, 3\}$ है। माना A पर, xRy यदि और केवल यदि $y = \max\{x, 1\}$ है, द्वारा परिभाषित एक संबंध R है। माना R में अवयवों की संख्या l है। माना R को स्वतुल्य और सममित बनाने के लिए इसमें जोड़े जाने वाले आवश्यक अवयवों की न्यूनतम संख्या क्रमशः m और n है। तो $l + m + n$ बराबर है :

(1) 12 (2) 14 (3) 11 (4) 13

Ans. Official answer NTA (1)

Sol. $A = \{-2,-1,0,1,2,3\}$

$R = \{(-2,1), (-1,1), (0,1), (1,1), (2,2), (3,3)\}$

$\ell = 6$

$m = 3$

$n = 3$

$\ell + m + n = 12$

Ex.9 Let A = {1, 2, 3}. The number of relations on A , containing (1, 2) and (2, 3), which are reflexive and transitive but not symmetric, is _____.

माना कि A = {1, 2, 3} है। A पर ऐसे संबंधों, जिनमें (1, 2) और (2, 3) हो, और जो स्वतुल्य और संक्रामक हों, परन्तु सममित नहीं हों, की संख्या है _____।

Ans. Official answer NTA (3)

Sol. A = {1, 2, 3}

Reflexive and transtive but not symmetric containing (1, 2) and (2, 3)

$R_1 = \{(1, 2) (2, 3) (1, 1) (2, 2) (3, 3) (1, 3)\}$

$R_2 = \{(1, 2) (2, 3) (1, 1) (2, 2) (3, 3) (3, 2)\}$

$R_3 = \{(1, 2) (2, 3) (1, 1) (2, 2) (3, 3) (2, 1)\}$

10. INVERSE OF A RELATION

Let A, B be two sets and let R be a relation from a set A to B. Then the inverse of R, denoted by R^{-1} , is a relation from B to A and is defined by $R^{-1} = \{(b, a) : (a, b) \in R\}$, Clearly, $(a, b) \in R \Leftrightarrow (b, a) \in R^{-1}$ Also, Dom of R = Range of R^{-1} and Range of R = Dom of R^{-1}

11. Number of Relations

$R : A \to A$ is a relation $n(A) = n$

1. Total number of relations = 2^{n^2}
2. Total number of reflexive relations = $2^{n(n-1)}$
3. Total number of symmetric relations = $2^{\frac{n(n+1)}{2}}$
4. Number of Equivalence Relations = B_n (Bell Number) (Refer extra reading material)